

Hospital Management and Information System

Final Year Project Report

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Project report in partial fulfillment of the requirements for the award of Bachelors
of Science (Computer Science) degree.

IN

DEPARTMENT OF COMPUTER SCIENCE

**SINDH MADARESSATUL ISLAM UNIVERSITY KARACHI,
PAKISTAN**

DECLARATION

We hereby declare that this project report is based on our original work except for citation and quotation which have been duly acknowledged. We also declare that it has not been previously and concurrently submitted for any other degree or award at Sindh Madressatul Islam University or any other institute.

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Hospital Management and Information System

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ABSTRACT

Hospitals manage large volumes of information related to patients, clinical encounters, inventory, and discharge processes. Manual record keeping and fragmented workflows often cause duplication, delays, and reduced traceability. This project presents a web-based Hospital Management Information System (HMIS) that centralizes patient registration, patient search, visit history, Medical Record entry, and discharge documentation within a secure role-based environment. The system also includes master data configuration (city, district, area, counters, and items) and an Inventory & Procurement Module that records Purchase Orders (PO) and Goods Received Notes (GRN) with automatic stock updates. The solution is implemented using Oracle APEX and Oracle Database to provide rapid development, strong data integrity, and centralized access for authorized users.

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In the name of Allah, the most Gracious and the Most Merciful.

Peace and blessing of Allah be upon Prophet Muhammad ﷺ

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DEDICATION

We dedicate this project to Almighty Allah, whose countless blessings and guidance enabled us to complete this work successfully. We also dedicate this project to our parents, whose prayers, support, and encouragement have been a constant source of strength throughout our academic journey. Finally, we dedicate this work to our respected supervisor and teachers for their valuable guidance, patience, and motivation, which played a vital role in the completion of this project.

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Chapter 1- Introduction

1.1 Introduction

The healthcare sector relies on timely, accurate, and secure information. Hospitals routinely handle patient demographics, visit information, clinical documentation, and discharge records. Manual processing can create duplication, delays, and incomplete audit trails.

An HMIS addresses these issues by digitizing routine workflows, centralizing data, and enabling role-based access to information. When implemented properly, an HMIS reduces paperwork and increases data quality by enforcing validations and standardized input.

In addition to clinical workflows, hospital operations depend on inventory availability. Procurement and receiving workflows ensure that essential supplies are available, and system-driven stock updates reduce errors in stock tracking.

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1.2 Project Overview

This Final Year Project implements a web-based HMIS using Oracle APEX and Oracle Database. The implemented solution includes patient registration and patient card generation, patient search, visit history, Medical Record entry, discharge summary generation, master data management, security administration, dashboard summaries, and an Inventory & Procurement Module.

The project emphasizes usability through structured forms and consistent navigation, and emphasizes security through RBAC. The Inventory & Procurement Module supports purchase orders and goods receipt processes and automatically updates stock when GRN entries are saved.

1.3 Problem Description

Hospitals using paper files or unconnected spreadsheets often face repeated entries and inconsistent data formats. Searching historical visit information can be slow and depends on manual filing quality. Administrative reporting is difficult when records are scattered.

Inventory tracking is also challenging in manual environments. Receiving goods without updating stock timely can lead to inaccurate availability, while missing audit trails reduce accountability.

The proposed HMIS provides a centralized, searchable platform with role-based access and standardized workflows to address these challenges.

1.4 Project Scope

- Patient Registration with patient card generation.
- Patient Search and profile viewing.
- Visit History viewing.
- Medical Record entry using tabbed UI.
- Discharge Summary creation and printing.
- Master Data Management: City, District, Area, Counter, Item.
- Security Administration: users, roles, access rights.
- Inventory & Procurement Module: PO and GRN workflows with automatic stock updates.
- Dashboard counters and KPI panels.

Out-of-scope: billing, pharmacy, laboratory modules, and external integrations; these are reserved for future work.

1.5 Chapter-wise Summary

Chapter 2 reviews literature, existing systems, and platform documentation. Chapter 3 specifies requirements and use cases. Chapter 4 presents design and database modeling. Chapter 5 documents the implementation with screenshots. Chapter 6 provides testing and evaluation. Chapter 7 concludes with future work proposals.

Chapter 2 – Literature Review

2.1 Introduction

This literature review summarizes knowledge relevant to developing a hospital management information system. The review includes commonly required HMIS modules, web-based architectures, security considerations, and platform guidance for Oracle APEX. Statements are supported using IEEE citations [1]–[8].

2.2 Books

Hospital information systems (HIS) integrate clinical and administrative processes, providing a shared repository for patient and operational data. Core benefits include reduced duplication, improved traceability, and faster access to patient history in time-sensitive environments.

Web-based hospital management information systems (HMIS) are widely adopted due to easier deployment and centralized maintenance. A browser-based interface reduces client installation overhead and supports access across workstations.

Common HMIS modules include patient registration, encounter/visit tracking, electronic clinical documentation, discharge documentation, and reporting. Supporting modules such as master data management and user administration improve data consistency and governance.

Role-Based Access Control (RBAC) is a practical approach for healthcare because responsibilities differ by role. Registration staff, clinicians, inventory staff, and administrators require different permissions, and RBAC enforces least-privilege access.

Procurement and inventory functions improve operational continuity. Recording purchase orders (PO) and goods received notes (GRN) creates an audit trail. Automatic stock updates on GRN reduce manual errors and improve availability visibility.

Oracle APEX is a database-centric development platform designed for data-driven web applications. Its request flow routes browser requests through ORDS to Oracle Database where the APEX engine processes metadata and executes logic close to the data [1].

Oracle APEX security features include application access control and authorization schemes that can be applied to pages and components, enabling fine-grained RBAC [2]–[4].

From a usability perspective, healthcare systems must reduce cognitive load. Staff benefit from standardized forms, consistent field labels, predictable navigation, and immediate validation feedback.

From a data quality perspective, master data (e.g., city/district/area, item lists) prevents inconsistent spelling and supports accurate reporting. Master data maintenance is an administrative responsibility in most HMIS deployments.

Research on HMIS implementation often emphasizes change management and training. Even a technically sound system may fail if users are not guided through the transition from paper workflows.

Security and privacy expectations can be addressed by strong authentication, authorization, and audit trails. While the current project focuses on RBAC, the platform can be extended with stronger auditing and encryption where required.

2.3 Articles

Industry articles consistently list patient registration, clinical records, discharge documentation, and inventory as core HMIS components because they are used daily. These resources also emphasize that dashboards and reporting tools help administrators monitor performance.

Articles also note that digitization reduces duplicate work and improves patient satisfaction by reducing waiting time and enabling better coordination among staff.

2.4 Applications and Existing Software Systems

Commercial HMIS products provide comprehensive modules but can impose licensing and customization costs. Smaller hospitals and academic projects often prefer modular or low-code approaches to reduce development time.

Open-source solutions may be adaptable but require careful security hardening and maintenance. In an academic setting, selecting a stable platform like Oracle APEX can reduce deployment complexity while enabling secure access control.

Comparing approaches highlights a gap: many prototypes describe features but lack strong governance over master data and procurement traceability. The proposed

system explicitly includes master data and procurement workflows to strengthen data consistency and operational control.

2.5 Websites and Platform Documentation

Oracle APEX platform documentation describes the three-tier request cycle and explains that APEX applications are metadata-driven and run inside Oracle Database [1]. This model benefits data-centric applications because business logic executes close to data.

Oracle APEX security documentation describes access control lists and authorization schemes for controlling access at application, page, and component levels [2]–[4]. These mechanisms can be used to implement RBAC for healthcare workflows.

Guides describing typical HMIS modules provide context for selecting which modules to implement first; patient and inventory workflows are frequently prioritized because they support day-to-day operations [5], [6].

2.6 Research Papers

Academic research on web-based HMIS describes improvements in data accessibility and coordination when systems are centralized. Many studies emphasize the importance of role-based access control to protect sensitive data.

Some studies focus on security mechanisms such as encryption and integrity checks, but even in systems without specialized security modules, applying RBAC and audit trails can reduce unauthorized access risks.

Published papers also report that structured documentation and discharge summaries improve continuity of care and support standardized processes.

In summary, the reviewed research supports adopting a web-based architecture, modular HMIS structure, and RBAC security, aligned with the approach used in this project.

2.7 Conclusion

The literature indicates that an HMIS should centralize patient workflows and operational processes while enforcing security controls and data integrity. Oracle APEX supports this through its database-centric architecture and built-in security components [1]–[4]. The identified module set and best practices guided the design and implementation of the proposed system.

Chapter 3 – Requirement Specification

3.1 Introduction

This chapter defines functional and non-functional requirements for the HMIS. Requirements are aligned with the implemented modules shown in the application screenshots. The purpose of this chapter is to provide a verifiable specification that supports design, implementation, and testing.

3.2 Proposed System

The proposed HMIS is a web-based system accessible via browser. It uses Oracle APEX for the application layer and Oracle Database for persistent storage.

The system implements patient registration, search, history, Medical Record entry, discharge summary, master data management, security administration, and inventory procurement workflows.

RBAC ensures users only access features required for their role, supporting privacy and operational governance.

3.3 Team Structure

The project follows a democratic team structure where members share responsibilities across analysis, development, and documentation. Collaboration improves resilience under deadlines and supports quality through peer review.

- Project Lead: planning, coordination, final documentation integration.
- System Analyst: requirements, use cases, validation of scope to implemented system.
- Database/Backend Developer: schema design, constraints, PL/SQL procedures and triggers.
- UI Developer: APEX pages, navigation, forms and reports, usability improvements.
- Tester: test plan, test cases, execution, defect tracking, evaluation summary.

3.4 Work Breakdown Structure (WBS)

WBS Code	Deliverable	Details
1.0	Planning	Scope, feasibility, documentation plan
2.0	Requirements	Functional/NFR, use cases, UI mapping

3.0	Design	Architecture, ERD, module decomposition
4.0	Implementation	APEX pages + DB schema + security configuration
5.0	Testing	Test cases, integration tests, UAT feedback
6.0	Report	Final report formatting, references, submission

3.5 Gantt Chart (Summary)

Week	Activities	Deliverables
1	Initiation	Scope + initial documentation
2	Requirements	SRS + use cases
3	Design	ERD + schema
4	Implementation-1	Masters + Patient Registration
5	Implementation-2	Security + Inventory
6	Testing	Test cases + bug fixes
7	Documentation	Screenshots + report draft
8	Finalization	Final report + references

3.6 Functional Requirements

ID	Module	Requirement
FR-01	Authentication	System shall authenticate users and create secure sessions.
FR-02	RBAC	System shall restrict menus/pages/actions based on assigned roles.
FR-03	Dashboard	System shall show counters/KPI panels for monitoring.
FR-04	Patient Registration	System shall register new patients and generate patient card.
FR-05	Patient Search	System shall provide search and listing of patients.
FR-06	Visit History	System shall display patient visit history.

FR-07	Medical Record	System shall allow entry and update of medical record data.
FR-08	Discharge Summary	System shall store and print discharge summaries.
FR-09	Master Data	System shall maintain City/District/Area/Counter/Item masters.
FR-10	Inventory & Procurement Module	System shall create purchase orders with line items.
FR-11	Inventory & Procurement Module	System shall create GRN and update stock automatically.

3.7 Non-Functional Requirements

ID	Attribute	Description
NFR-01	Security	Protect sensitive data through authentication and RBAC; restrict access to admin pages.
NFR-02	Usability	Consistent UI, clear labels, validation messages, minimal navigation steps.
NFR-03	Performance	Search and save should be responsive under normal load.
NFR-04	Reliability	Use constraints/validations to avoid inconsistent data states.
NFR-05	Maintainability	Consistent naming; modular pages; reusable LOVs and validations.
NFR-06	Availability	System should be available during operational hours; backups recommended.

3.8 Interface Requirements

User interfaces are delivered as web pages built in Oracle APEX. The application uses forms for data entry and interactive reports for search and listing.

The navigation menu and page access are controlled by authorization. Validation messages appear near fields to guide corrections.

The system supports printing/exporting of outputs such as patient card and discharge summary (as implemented in the UI).

3.9 Use Cases (Detailed Descriptions)

UC-01 – Login to HMIS

Primary Actor: Any authorized user

Preconditions: Valid credentials exist

Main Flow:

1. Open login
2. Enter credentials
3. System authenticates
4. Dashboard loads

Alternate Flow: Invalid credentials -> show error

Postconditions: Session active

Notes: This use case is validated against the implemented screen behavior shown in the project screenshots.

UC-02 – Register New Patient

Primary Actor: Registration Staff

Preconditions: Registration role assigned

Main Flow:

1. Open patient form
2. Enter data
3. Validate
4. Save
5. Generate card

Alternate Flow: Missing data -> block save

Postconditions: Patient saved

Notes: This use case is validated against the implemented screen behavior shown in the project screenshots.

UC-08 – Create GRN (Inventory & Procurement Module)

Primary Actor: Inventory Staff

Preconditions: PO exists and inventory role assigned

Main Flow:

1. Open GRN
2. Select PO
3. Enter received qty
4. Save GRN
5. Auto stock update

Alternate Flow: Invalid qty -> prompt

Postconditions: Stock updated

Notes: This use case is validated against the implemented screen behavior shown in the project screenshots.

Chapter 4 – System Design

4.1 Design Considerations

The system design balances usability, security, and data integrity. Hospital workflows must be fast and error-resistant, so the UI is designed around structured forms and consistent navigation.

Security is applied using RBAC. Administrative screens for roles and menu rights are restricted to administrators. Sensitive clinical screens are restricted to authorized clinical roles.

Data integrity is enforced through validations and relational constraints. Master data tables provide controlled vocabularies for location and items.

4.2 Architectural Design

The HMIS uses a layered architecture: presentation (browser), application logic (APEX processes and validations), and data (Oracle Database). ORDS acts as the HTTP gateway in the middle tier [1].

This architecture reduces complexity because the application is database-centric and minimizes additional middleware. Business rules can be implemented in PL/SQL and executed close to the data.

Authorization schemes and access control enable restricting pages and components according to roles [2]–[4].

4.3 Conceptual Design

Conceptual design defines major entities and relationships. Patients are linked to visits and medical records. Discharge summaries reference patients and, optionally, admission context. Inventory items are linked to procurement documents (PO and GRN).

Master data entities (city, district, area) support consistent address data. Security entities (users, roles) define access control.

The ERD provides a visual representation of these relationships and guided schema development.

4.4 Physical Design

Physical design focuses on implementation details such as table definitions, indexes, constraints, and triggers. Primary keys uniquely identify each record, and foreign keys enforce relationships.

For procurement, GRN processing updates item stock. This can be implemented through database triggers or page processes that adjust CURRENT_STOCK based on received quantity.

Proper indexing on patient identifiers and names improves search performance.

4.5 High-Level Design (HLD)

- Authentication + RBAC module
- Dashboard module
- Master Data module
- Patient module (registration/search/history)
- Medical Record module
- Discharge Summary module
- Inventory & Procurement Module (PO/GRN/Stock Update)
- Security Administration module

Each module is implemented as a set of Oracle APEX pages. Shared Components are used for common items such as List of Values (LOVs), authorization schemes, and navigation menus.

The database layer supports these modules through normalized tables and controlled relationships.

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The database layer supports these modules through normalized tables and controlled relationships.

4.6 Low-Level Design (LLD)

At the page level, APEX forms perform CRUD operations (Create, Read, Update, Delete) using automatic row processing or custom PL/SQL processes. Validations enforce mandatory fields and business rules.

For patient registration, validations ensure required demographic fields are not empty. For procurement, validations ensure quantities are positive and that received items exist in the item master.

RBAC is implemented by attaching authorization schemes to pages, buttons, and regions to hide or block access based on roles.

4.7 Database Schema

The following data dictionary summarizes major tables used by the HMIS. Field lists are representative of typical implementations and align with the screens shown in the project.

4.7.1 Patient Table Fields

Field	Type	Null?	Description
PATIENT_ID	NUMBER	No	Primary key; unique patient identifier
FULL_NAME	VARCHAR2	No	Patient full name
FATHER_NAME	VARCHAR2	Yes	Father/guardian name
CNIC	VARCHAR2	Yes	National ID (if applicable)
DOB	DATE	Yes	Date of birth
GENDER	VARCHAR2	Yes	Gender
MOBILE	VARCHAR2	Yes	Contact number
ADDRESS	VARCHAR2	Yes	Address details
CITY_ID	NUMBER	Yes	FK to CITY
DISTRICT_ID	NUMBER	Yes	FK to DISTRICT
AREA_ID	NUMBER	Yes	FK to AREA
CREATED_AT	DATE	No	Record creation timestamp

4.7.2 Item Table Fields

Field	Type	Null?	Description
ITEM_ID	NUMBER	No	Primary key; unique item identifier

ITEM_NAME	VARCHAR2	No	Item name
UOM	VARCHAR2	Yes	Unit of measure
CATEGORY	VARCHAR2	Yes	Item category
REORDER_LEVEL	NUMBER	Yes	Low-stock threshold
CURRENT_STOCK	NUMBER	No	Current stock quantity
STATUS	VARCHAR2	No	Active/Inactive

4.7.3 Purchase Order Header Fields

Field	Type	Null?	Description
PO_ID	NUMBER	No	Primary key
PO_DATE	DATE	No	Purchase order date
SUPPLIER_NAME	VARCHAR2	Yes	Supplier name
STATUS	VARCHAR2	No	Draft/Approved/Closed
TOTAL_AMOUNT	NUMBER	Yes	Computed total

4.7.4 GRN Header Fields

Field	Type	Null?	Description
GRN_ID	NUMBER	No	Primary key
PO_ID	NUMBER	Yes	Reference purchase order
GRN_DATE	DATE	No	Goods receiving date
RECEIVED_BY	VARCHAR2	Yes	Receiver user/staff
REMARKS	VARCHAR2	Yes	Notes

4.8 Entity Relationship Diagram (ERD)

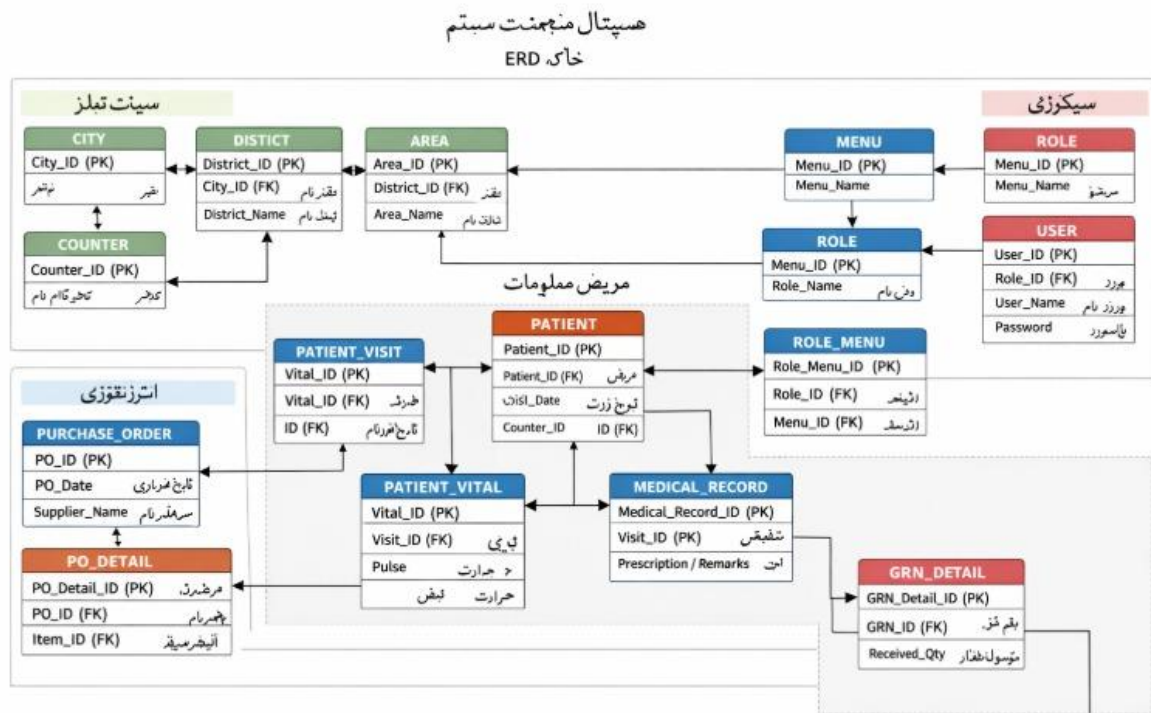


Figure 4.1: Entity Relationship Diagram (ERD) of HMIS Database

Chapter 5 – System Implementation

5.1 Development Environment and Tools

The HMIS is developed in Oracle APEX, which provides form and report components for rapid development. Oracle Database stores master and transactional data.

PL/SQL is used for server-side validations and processes. Shared Components are used to manage navigation, authorization schemes, and LOVs.

Development activities include page creation, process/validation configuration, role assignment, and iterative testing.

5.2 Hardware and Software Specifications

Category	Specification
Development Hardware	Laptop/PC with 8GB+ RAM recommended
Operating System	Windows/Linux
Platform	Oracle APEX with ORDS
Database	Oracle Database
Client	Modern web browser (Chrome/Edge/Firefox)

5.3 Deployment and Configuration

The application is deployed on an Oracle APEX environment and accessed via browser. ORDS receives HTTP requests and routes them to the database, where APEX renders pages.

User accounts and roles are configured through Security Administration. Admin assigns roles and menu permissions. Users must re-login for role changes to take effect.

Project URL for demonstration: <https://oracleapex.com/ords/r/aptf/hospital-management-information-system/home> (accessed Jan. 31, 2026).

5.4 Module-wise Implementation (Screenshots and Explanation)

5.3.1 Login & Role-Based Access Control

The login page validates user credentials. On successful login, a session is created and the user is redirected to the dashboard. Failed login attempts are rejected with an error message.

RBAC is applied after login. The navigation menu and page access depend on roles assigned to the user.

Security is strengthened by limiting administrative pages to admin roles only.

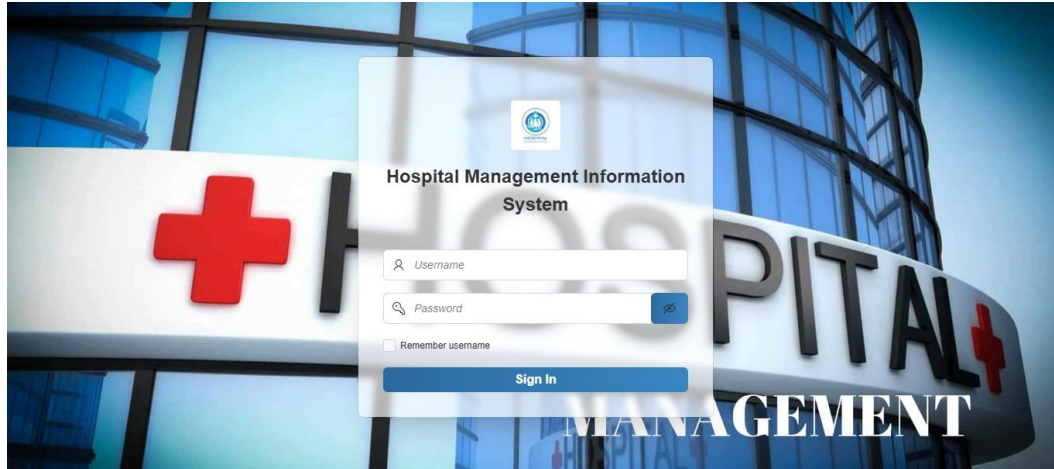


Figure 5.1: User Login Screen

5.3.2 Security Administration

Security Administration includes screens for roles, users, and access rights. Roles define privileges for modules, and users are mapped to roles.

Menu/Access rights configuration determines which pages a role can open. Authorization schemes implement these checks.

Administrative actions should be performed carefully to maintain least-privilege access.

Master Setups

Reports

Security

Menu Setup

Role Setup

User Setup

Transaction

Hospital Management Information System

sami

SECURITY MENU SETUP

Q

Go

Actions

Create

	Menu Parent ID	Menu Name	Page No	Create By	Create Date	Update By	Update Date	Menu Icon	Form Page No
		Security			12/3/2025				
	19	Purchase Order	129		1/12/2026				130
	1	ITEM SETUP	117		12/8/2025				118
	19	GRN APPROVAL	136		1/16/2026				
	1	DISTRICT SETUP	103		12/8/2025				104
	2	PATIENT REGISTRATION	107		12/8/2025				108
	3	Menu Setup	123		1/9/2026				124
	1	COUNTER	109		12/8/2025				110
	2	SEARCH PATIENT	113		12/8/2025				
	1	CITY SETUP	101		1/9/2026				102
	2	Inventory			1/12/2026				

Figure 5.2: Menu / Access Rights Configuration

Role Description	Create By	Create Date	Update By	Update Date
RECEPTIONIST		1/20/2026		
DOCTOR		12/8/2025		
NURSE		1/20/2026		
USER		12/8/2025		
ADMIN		12/9/2025		

Figure 5.3: Role Management Screen

Role Description

Figure 5.4: User Role Assignment Screen

5.3.3 Navigation & Modules

This module is implemented using APEX forms and reports. Validations ensure data quality and consistent usage.

Navigation and visibility are controlled through authorization schemes where applicable.

Screenshots below provide evidence of the implemented user interface.

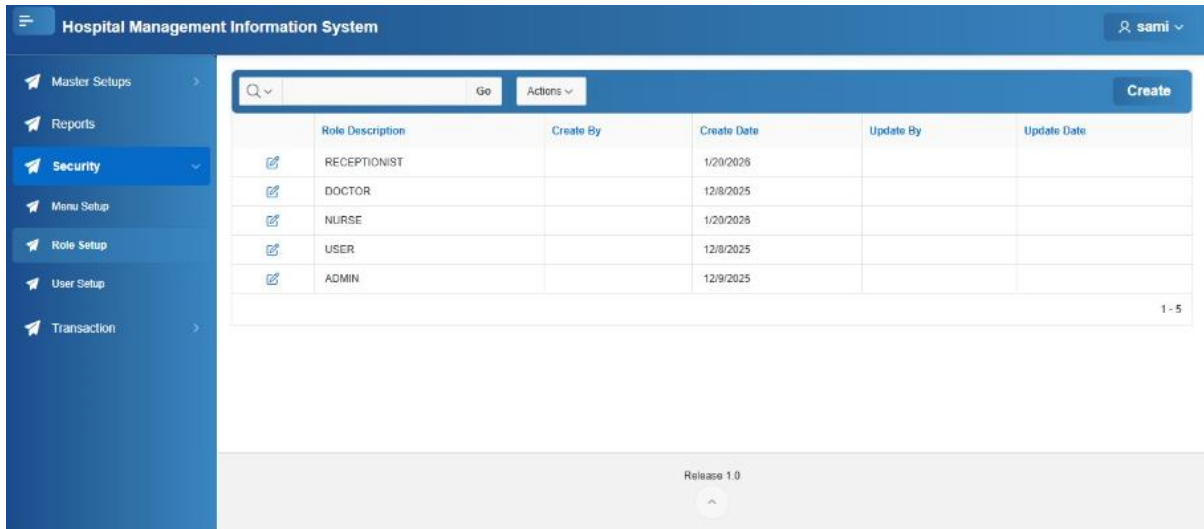


Figure 5.5: Application Navigation / Modules Menu

5.3.4 Dashboard

This module is implemented using APEX forms and reports. Validations ensure data quality and consistent usage.

Navigation and visibility are controlled through authorization schemes where applicable.

Screenshots below provide evidence of the implemented user interface.

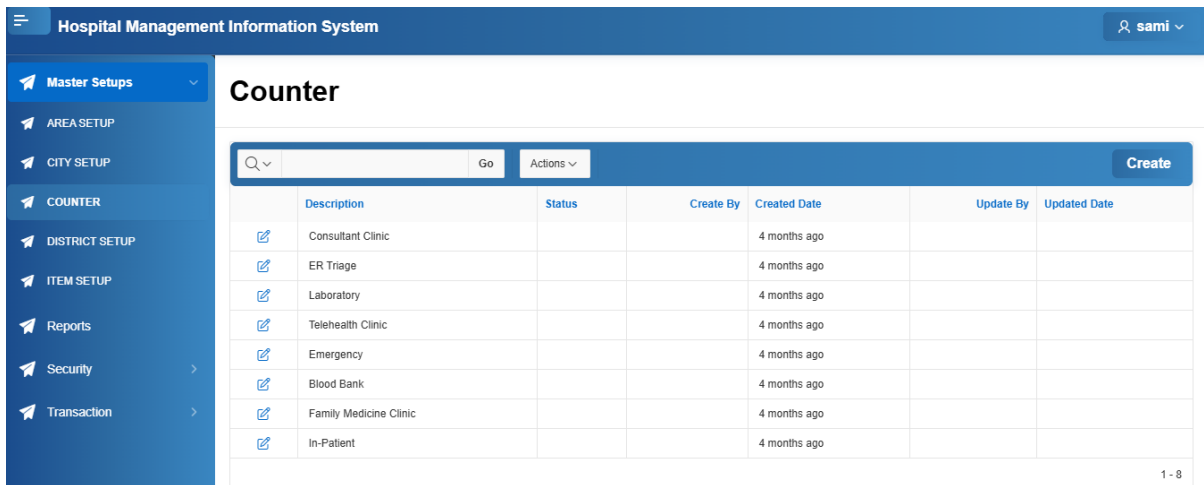


Figure 5.6: Dashboard Counters Overview

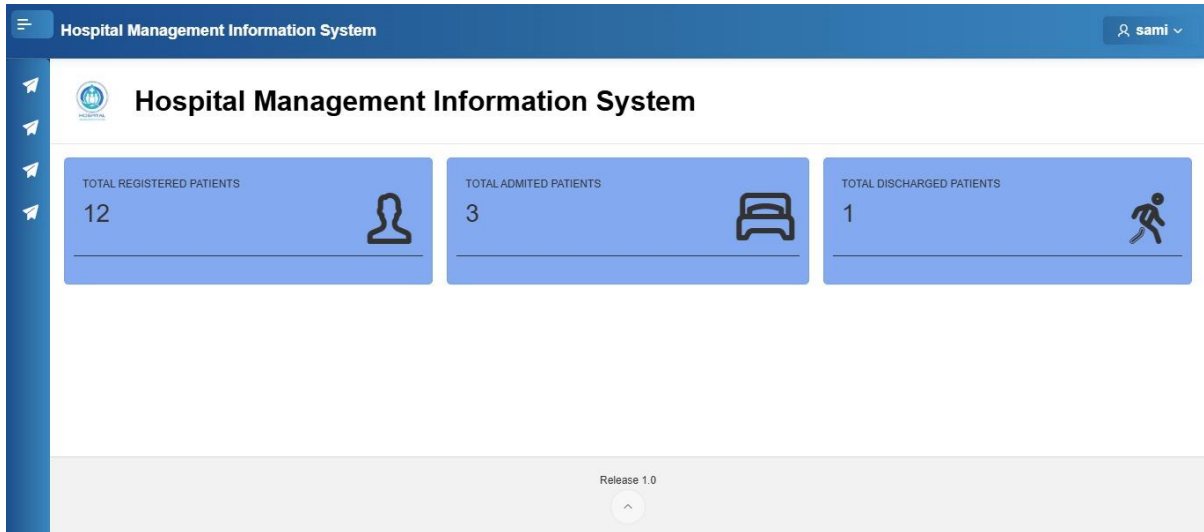


Figure 5.7: Dashboard KPI Panels

5.3.5 Master Data Management

Master data pages maintain City, District, Area, Counter, and Item masters. These masters are referenced by transactional modules to reduce inconsistencies.

Uniqueness validations prevent duplicates. Inactive flags can be used to avoid deleting master records that are referenced by existing transactions.

Item master includes current stock and reorder levels to support inventory monitoring.

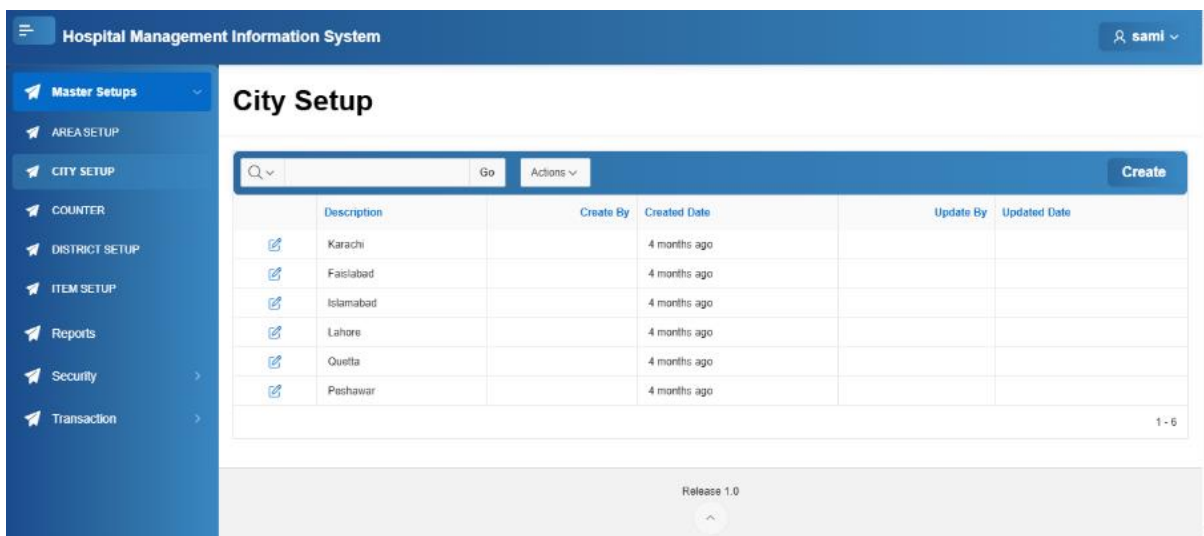


Figure 5.8: City Master Data Screen

Hospital Management Information System							
sami							
Master Setups AREA SETUP CITY SETUP COUNTER DISTRICT SETUP ITEM SETUP Reports Security Transaction							
Distrtrict Setup							
Q Go Actions Create							
	Description	City Id Fk	Create By	Created Date	Update By	Updated Date	
	Korangi	1		4 months ago			
	Kemari	1		4 months ago			
	Karachi East	1		4 months ago			
	Malir	1		4 months ago			
	Karachi South	1		4 months ago			
	Karachi Central	1		4 months ago			
	Karachi West	1		4 months ago			
							1 - 7

Figure 5.9: District Master Data Screen

Hospital Management Information System							
sami							
Master Setups AREA SETUP CITY SETUP COUNTER DISTRICT SETUP ITEM SETUP Reports Security Transaction							
Area Setup							
Q Search: All Text Columns Go Actions Create Reset							
	Description	City Id Fk	District Id Fk	Create By	Created Date	Update By	Updated Date
	New Karachi 11-F	Karachi	Karachi South		1/7/2026		
	Korangi#1	Karachi	Korangi		9/21/2025		
	Korangi#2	Karachi	Korangi		9/22/2025		
	Korangi#5	Karachi	Korangi		9/22/2025		
	Korangi#3	Karachi	Korangi		9/22/2025		
	Korangi#4	Karachi	Korangi		9/22/2025		
1 rows selected							Total 6

Figure 5.10: Area Master Data Screen

Hospital Management Information System							
sami							
Master Setups							
AREA SETUP							
CITY SETUP							
COUNTER							
DISTRICT SETUP							
ITEM SETUP							
Reports							
Security							
Transaction							

Item Setup							
Go Actions Create							
	Item Description	Item Category	Item Rate	Create By	Created Date	Update By	Updated Date
	Paracetamol 500mg Tablet	M	10	1	3 months ago		
	Amoxicillin 500mg Capsule	M	15	1	3 months ago		
	Azithromycin 250mg Tablet	M	25	1	3 months ago		
	Cefixime 200mg Tablet	M	20	1	3 months ago		
	Cetirizine 10mg Tablet	M	5	1	3 months ago		
	Pantoprazole 40mg Tablet	M	12	1	3 months ago		
	Omeprazole 20mg Capsule	M	10	1	3 months ago		
	Ibuprofen 400mg Tablet	M	8	1	3 months ago		
	Metformin 500mg Tablet	M	7	1	3 months ago		
	Losartan 50mg Tablet	M	10	1	3 months ago		
	Diclofenac Sodium 50mg Tablet	M	6	1	3 months ago		

Figure 5.11: Item Master / Inventory Items Screen

5.3.6 Patient Registration

Patient registration captures demographics, contact details, and address references. Mandatory fields are validated to ensure completeness.

After saving, the system generates a patient identifier and provides a patient card/summary. This supports quick identification and future search.

Consistent use of master data for location improves reporting and reduces errors.

Hospital Management Information System							
sami							
Master Setups							
Reports							
Security							
Transaction							

Patient Medical Record							
Save Discharge Cancel							
Mr Id Fk 1	VISIT_HISTORY_ID_FK 2	Mr Name MR ASIF S/O ANWER	DOB 9/9/2002	Age 23			
Mr Visitdate 1/26/2026	Arrival Date 1/28/2026	Departure Datetime					

Patient Vital	Diagnosis	Final Diagnosis	Medication	Laboratory	Summary Remarks		
Go Actions							
Vital Date	Vital Bp Sis	Vital Dys	Vital Temp	Vital Pulse	Vital Res Rate	Vital Height	Vital Weight
1/26/2026	100	100	100	100	100	10	100

1 - 1

PATIENT REGISTRATION CARD
✕

AL-SHIFA HOSPITAL

Patient Registration Card

MR Code:

2601-00011

Registration Date:

20-JAN-2026

Patient Name:

MR Abdullah Iftikhar Ahmed

Date of Birth:

1/27/2026

Gender:

MALE

Address:

A-102 Block C North Nazimabad

This card is system generated - No signature required.

Figure 5.12: Patient Registration Form and Patient Card Generation

5.3.7 Patient Search

This module is implemented using APEX forms and reports. Validations ensure data quality and consistent usage.

Navigation and visibility are controlled through authorization schemes where applicable.

Screenshots below provide evidence of the implemented user interface.

Hospital Management Information System
🔍 sami ▾

📁 Master Setups >
📁 Reports
📁 Security >
Transaction ▾
📁 Inventory >
📁 PATIENT MEDICAL RECORD
PATIENT REGISTRATION
📁 Patient Visit Detail
📁 SEARCH PATIENT

Patient Registration

Q ▾

Go

Actions ▾

Create

	Print Slip	Mr Code	Mr Reg Date	Mr Title	Mr Name	Mr Sdw	Mr Sdw Name	Mr Dob	Mr Gender	Mr Address	District Id Fk	City Id Fk	Area Id Fk	Mr Flat No	Mr Phone No	Mr Mobile
		2509-00003	9/24/2025	1	Umer	S	AFTAB	9/4/2019	M	HOUSE # 677	1	1	2		0333892929	
		2601-00012	1/20/2026	1	Areeb	s	Ashraf	9/2/2000	M	House # 6-3/337	5	1			0321029292	03210292
		2510-00005	10/4/2025	4	AMINA	D	SHAREEF	10/1/2025	F	HOUSE # 456			3		03042208689	03042208
		2510-00006	10/4/2025	2	AMINA	W	SALHUDDIN	10/11/1994	F	HOUSE # 67	2	1			03178976378	03178976
		2509-00001	9/24/2025	1	ASIF	S	ANWER	9/9/2002	M	HOUSE # 456	1	1	1		0383828892	
		2510-00007	10/22/2025	1	FARHAN	S	AMAN	10/8/2014	M	HOUSE # 456	1	1	1	323	03456728919	03456728

Hospital Management Information System | sami

Search Patient

Enter Mr No : 2601-00011

Mr Id Pk	Mr Name	Mr Dob	Age	Mr Phone No
11	MR Abdullah Iftikhar Ahmed	1/27/2026	0	03087573499

Release 1.0

Figure 5.13: Search Patient / Patient Listing

5.3.8 Patient History

This module is implemented using APEX forms and reports. Validations ensure data quality and consistent usage.

Navigation and visibility are controlled through authorization schemes where applicable.

Screenshots below provide evidence of the implemented user interface.

Hospital Management Information System | sami

Patient Visit History

Patient Name: MR ASIF S/O ANWER

Dob: 9/9/2002 | Age: 23 | Phone No: 0303028892 | Visitdate: 1/31/2026

Status: Visited | Counter: [Dropdown]

Save **Cancel**

Visit History Id Pk	P Name	Status	Mr Visitdate	Created Date
2	MR ASIF S/O ANWER	Visited	1/26/2026	1/26/2026
1	MR ASIF S/O ANWER	Discharge	1/23/2026	1/23/2026

1 - 2

Figure 5.14: Patient Visit History Screen

5.3.9 Medical Record

The Medical Record screen provides a tabbed interface for clinical documentation. Tabs group related fields to keep the interface organized.

Clinical staff can enter notes, diagnosis, and other encounter information. Save operations commit the data to the database.

Access to the Medical Record screen is restricted to authorized clinical roles.

Hospital Management Information System | sami

GOOD RECIEVE NOTED

Gm Date: 1/23/2026 | PURCHASE ORDER ID: 2601-00001 / ARMAN ALI / 23-Jan-2026

Remarks: test | Status: Approved

Cancel

Search: All Text Columns | Go | Actions | Edit | **Save** | Add Row | Reset

	Item	Batch No	PO Qty	Rec Qty	Bal Qty	GRN Qty
☒		12345	100	0	100	100
☐		124578	110	0	110	110
☐		6655	150	0	150	150

1 rows selected | Total 3

Figure 5.15: Patient Medical Record (Tabbed View)

5.3.10 Discharge Summary

This module is implemented using APEX forms and reports. Validations ensure data quality and consistent usage.

Navigation and visibility are controlled through authorization schemes where applicable.

Screenshots below provide evidence of the implemented user interface.

PATIENT DISCHARGE SUMMARY

Patient Discharge Summary

AL-SHIFA HOSPITAL

Patient Discharge Summary

Patient Information

Patient Name:	ASIF	Arrival Date:	23-JAN-2026
MR Code:	2509-00001	Departure Date:	26-JAN-2026
Date of Birth:	09-SEP-2002	Phone No:	0383828892
Address:	HOUSE # 456		

Diagnosis

Final Diagnosis

Summary

Figure 5.16: Discharge Summary Form

5.3.11 Inventory & Procurement Module

The procurement workflow begins with Purchase Order creation. Users add line items and quantities. The PO acts as the planned procurement record.

When items are received, the user creates a GRN, selects the PO, and enters received quantities. Saving the GRN updates CURRENT_STOCK automatically.

Automatic stock update reduces manual calculation errors and provides current availability for other processes.

Hospital Management Information System

sami

Master Setups

Reports

Security

Transaction

Purchase Order

Code: 2601-00002

Date: 1/29/2026

Supplier: Abdullah

Remarks: Approved

Status: Pending

Cancel

Search: All Text Columns

Go

Actions

Edit

Save

Add Row

Reset

	Item	Item Rate	PO Rate	Quantity	Amount
☒	Augmentin Syrup 156mg/5ml	150	150	15	2250
☒	Pantoprazole 40mg Tablet	12	15	100	1500

4 rows selected

Total 4

Figure 5.17: Purchase Order Screen

Hospital Management Information System

sami

Master Setups

AREA SETUP

CITY SETUP

COUNTER

DISTRICT SETUP

ITEM SETUP

Reports

Security

Transaction

Item Setup

Search:

Go

Actions

Create

	Item Description	Item Category	Item Rate	Create By	Created Date	Update By	Updated Date
	Paracetamol 500mg Tablet	M	10	1	3 months ago		
	Amoxicillin 500mg Capsule	M	15	1	3 months ago		
	Azithromycin 250mg Tablet	M	25	1	3 months ago		
	Cefixime 200mg Tablet	M	20	1	3 months ago		
	Cetirizine 10mg Tablet	M	5	1	3 months ago		
	Pantoprazole 40mg Tablet	M	12	1	3 months ago		
	Omeprazole 20mg Capsule	M	10	1	3 months ago		
	Ibuprofen 400mg Tablet	M	8	1	3 months ago		
	Metformin 500mg Tablet	M	7	1	3 months ago		
	Losartan 50mg Tablet	M	10	1	3 months ago		
	Diclofenac Sodium 50mg Tablet	M	6	1	3 months ago		

Figure 5.18: Purchase Order Details / Line Items

Hospital Management Information System
sami

Master Setups
Reports
Security
Transaction

GOOD RECIEVE NOTED

Gm Date
1/23/2026

PURCHASE ORDER ID
2601-00001 / ARMAN ALI / 23-Jan-2026

Remarks
test

Status
Approved

Cancel

Search: All Text Columns
Go
Actions
Edit
Save
Add Row
Reset

	Item	Batch No	PO Qty	Rec Qty	Bal Qty	GRN Qty
☒		12345	100	0	100	100
☐		124578	110	0	110	110
☐		6655	150	0	150	150

1 rows selected
Total 3

Figure 5.19: Goods Received Note (GRN) Screen (Auto Stock Update)

5.5 Coding Standards and Practices

Although Oracle APEX is low-code, consistent naming is important. Page names, items, and database objects follow clear naming conventions.

Validation rules are centralized where possible and reused across pages. LOVs are defined in Shared Components for consistent selection lists.

Changes are tested in development environment before being demonstrated in the deployed instance.

Chapter 6 – System Testing & Evaluation

6.1 Testing Strategy

Testing verifies that the system meets functional requirements and behaves reliably under expected use. Test cases include both positive and negative scenarios.

RBAC testing ensures unauthorized roles cannot access restricted pages. Inventory tests verify that stock updates correctly on GRN save.

Integration testing ensures that patient workflows connect: registration → search → history → medical record → discharge.

6.2 Testing Levels

- Unit Testing: validations, computations, and CRUD per screen.
- Integration Testing: data flows across modules.
- System Testing: end-to-end scenarios with multiple roles.
- User Acceptance Testing: usability feedback during demonstrations.

6.3 Test Cases

Test Case ID	Module	Scenario	Expected Result	Status
TC-01	Login	Valid username/password	User logged in; session started	Pass
TC-02	Login	Invalid credentials	Error message; session not created	Pass
TC-03	RBAC	User without Admin role opens Admin page	Access denied / page hidden	Pass
TC-04	Patient Registration	Create new patient with required fields	Patient saved; ID generated	Pass
TC-05	Patient Registration	Try save with missing mandatory name	Validation error; not saved	Pass
TC-06	Patient Search	Search by name fragment	Matching list shown	Pass
TC-07	Patient History	Open Visit History for patient	Visit list appears	Pass
TC-08	Medical Record	Enter Medical Record details and save	Record saved and linked to patient	Pass

TC-09	Discharge Summary	Save discharge summary	Printable discharge generated	Pass
TC-10	Master Data	Add City/District/Area	Record saved and available	Pass
TC-11	Inventory & Procurement Module	Create PO with items	PO saved with line items	Pass
TC-12	Inventory & Procurement Module	Create GRN for PO	GRN saved	Pass
TC-13	Inventory & Procurement Module	Verify stock update after GRN	Item CURRENT_STOCK increases by received qty	Pass
TC-14	Inventory & Procurement Module	Attempt GRN with invalid quantity (if constrained)	System warns/blocks entry	Pass
TC-15	Security Admin	Assign role to user	User gains access after relogin	Pass
TC-16	Security Admin	Remove role from user	User access removed	Pass

6.4 Test Execution Summary

All test cases were executed with sample data in the APEX environment. Defects were logged and resolved iteratively.

Most defects were related to validation rules and required field checks. These were resolved by adjusting APEX validations and database constraints.

Inventory tests confirmed that saving a GRN increases CURRENT_STOCK for the received items automatically.

RBAC tests confirmed that admin-only pages are hidden or blocked for non-admin roles.

6.5 Evaluation

Usability evaluation indicates that structured forms and consistent layout reduce learning time. Search-based access to patient records improves retrieval speed.

Security evaluation indicates that RBAC effectively restricts access; however, future work can enhance auditing and monitoring.

Performance evaluation is acceptable for typical operations in an academic environment; indexing and query tuning can further improve search responsiveness.

Chapter 7 – Conclusion

7.1 Key Achievements

- Implemented a centralized HMIS with patient registration, search, visit history, Medical Record, and discharge summary.
- Implemented master data management for city/district/area/counter/item.
- Implemented Inventory & Procurement Module with PO, GRN and automatic stock update.
- Implemented security administration and RBAC.
- Prepared test plan and executed test cases.

7.2 Significance and Impact

The system demonstrates how low-code platforms can deliver practical hospital workflows with strong data integrity and security controls.

Centralized patient records improve continuity of care by making historical information accessible to authorized staff.

Procurement traceability and automatic stock updates improve operational readiness and reduce the risk of stock-outs.

The project provides a foundation for adding additional modules such as billing, pharmacy, and laboratory in future versions.

7.3 Limitations

The current system implements the modules demonstrated in the UI screenshots. Some advanced hospital functions are not included due to scope limitations.

The report includes conceptual database table lists where exact schema names may differ. Future versions can include a direct export of schema definitions for complete accuracy.

Automated reporting and analytics can be expanded in future work.

7.4 Future Work

- Billing and invoicing module with receipts.
- Pharmacy module with dispensing, expiry and interactions.
- Laboratory module with test ordering and report upload.
- Advanced dashboards and KPI analytics.
- Improved audit trails and activity logs.
- Dedicated deployment with backups and monitoring.

7.5 Summary of Learning Outcomes

The project improved understanding of requirements engineering, modular design, and database modeling.

The team gained hands-on experience with Oracle APEX development, RBAC configuration, and database-driven workflows.

Documentation and testing activities enhanced professional communication and quality assurance skills.

APPENDIX A – User Manual

A.1 Login

Open the HMIS URL in a browser. Enter username and password. Click Login. If credentials are correct, the dashboard loads and menus appear based on the assigned role.

Tips: Follow validation messages. Ensure you have the required role to access the module. Use search to quickly locate records.

A.1 Patient Registration

Navigate to Patient Registration. Enter demographic details. Select City/District/Area if configured. Save the form. Use the patient card output/print option if available.

Tips: Follow validation messages. Ensure you have the required role to access the module. Use search to quickly locate records.

A.1 Patient Search and History

Navigate to Patient Search. Search by ID or name. Open patient profile. Navigate to Visit History to review previous visits.

Tips: Follow validation messages. Ensure you have the required role to access the module. Use search to quickly locate records.

A.1 Medical Record

From the patient context, open Medical Record. Use tabs to enter complaints/diagnosis/notes. Save changes. Verify record is visible on revisit.

Tips: Follow validation messages. Ensure you have the required role to access the module. Use search to quickly locate records.

A.1 Discharge Summary

Open Discharge Summary. Enter discharge notes and follow-up instructions. Save and print/export discharge summary.

Tips: Follow validation messages. Ensure you have the required role to access the module. Use search to quickly locate records.

A.1 Master Data

Admin users maintain City/District/Area and Item masters. Add or edit records carefully to avoid duplicates.

Tips: Follow validation messages. Ensure you have the required role to access the module. Use search to quickly locate records.

A.1 Inventory & Procurement

Open Purchase Order screen. Add line items and save PO. When goods arrive, open GRN, select PO, enter received quantities, and save. Confirm item stock updated automatically.

Tips: Follow validation messages. Ensure you have the required role to access the module. Use search to quickly locate records.

APPENDIX B – Project Screenshots Index

This appendix lists the screenshots included in Chapter 5 as evidence of implementation.

Figure	Description
5.1	User Login Screen
5.2	Menu / Access Rights Configuration
5.3	Role Management Screen
5.4	User Role Assignment Screen

5.5	Application Navigation / Modules Menu
5.6	Dashboard Counters Overview
5.7	Dashboard KPI Panels
5.8	City Master Data Screen
5.9	District Master Data Screen
5.10	Area Master Data Screen
5.11	Item Master / Inventory Items Screen
5.12	Patient Registration Form and Patient Card Generation
5.13	Search Patient / Patient Listing
5.14	Patient Visit History Screen
5.15	Patient Medical Record (Tabbed View)
5.16	Discharge Summary Form
5.17	Purchase Order Screen
5.18	Purchase Order Details / Line Items
5.19	Goods Received Note (GRN) Screen (Auto Stock Update)

APPENDIX C – Project Link

Project demonstration link: <https://oracleanapex.com/ords/r/aptf/hospital-management-information-system/home> (accessed Jan. 31, 2026).

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